

Cycle Structure Generation

All Possible Cycle Structures For The Full ERG Chain Subset from Baca Dataset

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#1 Understanding The Data At Hand

1	Individual	Number	Breakpoint 1 chromosome	Breakpoint 1 strand	Breakpoint 1 position	Breakpoint 2 chromosome	Breakpoint 2 strand	Breakpoint 2 position
1584	P08-501	4	21	+	39887956	21	-	42873929
1589	P08-501	11	1	-	179143829	21	+	48033460

Breakpoint Chromosome Range: 1 - 24

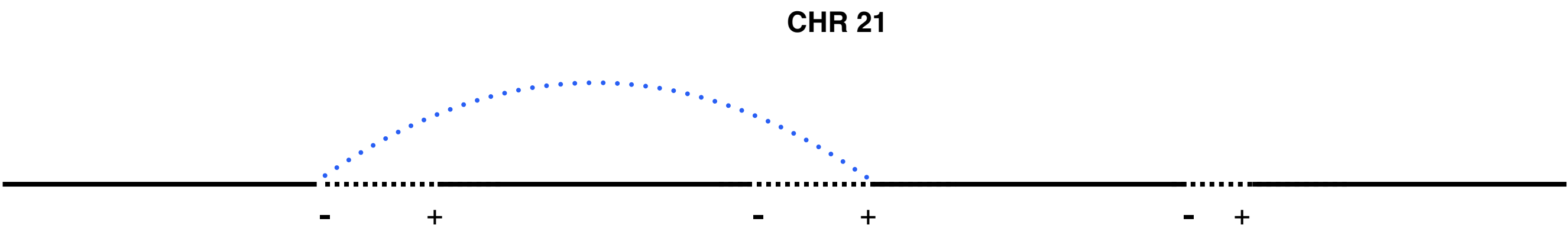
Breakpoint Location: Location of the Breakpoint in Base Pairs (bp)

Breakpoint Strand: “-” (Left Node of Breakpoint) & “+” (Right Node of Breakpoint)

Number of DSBs = $2 \times n_rejoins = 2 \times$ (Total Rows for Patient Y)

Total Possible Cycle Structures = $(n_rejoins - 1)!!$ —> Double Factorial

What does each Row say?

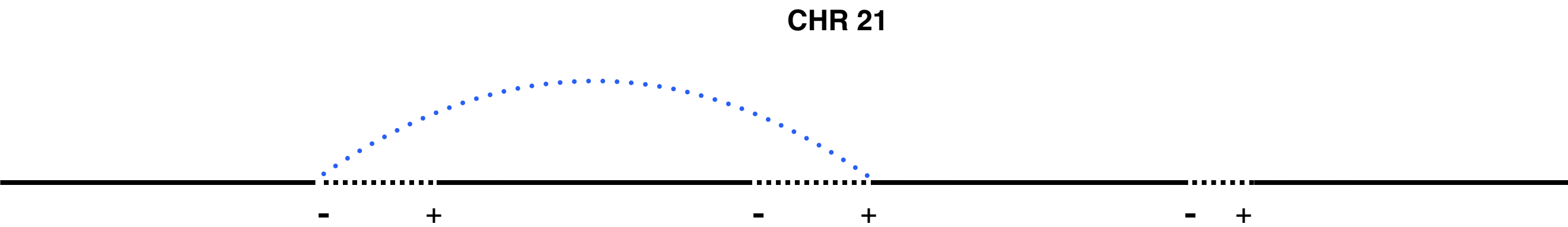


Each Row is **ONE** rejoin. A row for this example would look something like this:

Patient ID	Breakpoint 1 Chromosome	Breakpoint 1 Location	Breakpoint 1 Strand	Breakpoint 2 Chromosome	Breakpoint 2 Location	Breakpoint 2 Strand
P0X - 10XX	21	392XXXX	-	21	458XXXX	+

#2 Compiling Required Data for Cycle Enumeration

Let's agree on this



- 1. If each row in the table shows 1 rejoin. Then it shows 2 DSBs for THAT rejoin.
- 2. EACH BREAKPOINT Location has 2 strands. One “-” (Left Side) & One “+” (Right Side)

Patient ID	Breakpoint 1 Chromosome	Breakpoint 1 Location	Breakpoint 1 Strand	Breakpoint 2 Chromosome	Breakpoint 2 Location	Breakpoint 2 Strand
P08-501	21	39887956	+	21	42873929	-
P08-501	1	179143829	-	21	48033460	+

In the above example, there are 4 Total Breakpoints.

Each breakpoint would have a + and - strand, which then makes the total chain data look like this:

```
(chr1, 179143829, '+') (chr1, 179143829, '-')
(chr21, 39887956, '+') (chr21, 39887956, '-')
(chr21, 42873929, '+') (chr21, 42873929, '-')
(chr21, 48033460, '+') (chr21, 48033460, '-')
8 nodes total (= 2 x n_DSBs).
```

Rearrangement Edges (Rows of Data)

Patient ID	Breakpoint 1 Chromosome	Breakpoint 1 Location	Breakpoint 1 Strand	Breakpoint 2 Chromosome	Breakpoint 2 Location	Breakpoint 2 Strand
P08-501	21	39887956	+	21	42873929	-
P08-501	1	179143829	-	21	48033460	+

For Patient **P08-501**,
The Rearrangement Data would look something like this:

```
Row 4: (chr21, 39887956, '+') ↔ (chr21, 42873929, '-')
Row 11: (chr1, 179143829, '-') ↔ (chr21, 48033460, '+')
```

Reference Edges (Building from BPs)

For this patient, **Chr 21** has **3 DSBs**.



But the only data we are given, is about REJOINS. That too, for specific strands. The red marked strands have NO DATA provided for them. Meaning they did not participate in the rejoin.

Notice the black edges (reference edges) are removed because the data currently ONLY tells us about BPs, but how can we use them to know REFERENCE EDGES?

Reference Edges (Building from BPs)

Locations for the BPs are linear. Meaning

HIGHER BP Location Number = **FARTHER** in the Chromosome

This gives us an intuition about where the breakpoints are next to each other. The below diagram shows BPs in INCREASING ORDER of appearance.



Using this we know, that BP 1 is to the left of BP 2 , and so on.

To create reference edges, this approach is used:

- 1. Sort BPs in Increasing Order.
- 2. Set Open Ends (Telomeres):
 - 1. LOWEST BP location means it is the LEFTMOST BP.
 - 2. “-” strand of that BP is the LEFT MOST NODE which does NOT have any rejoin.
 - 3. It automatically becomes a telomere edge since NOTHING is to the LEFT of “-” strand.



Reference Edges (Building from BPs)

- 3. For the remaining strands (“+”)
 - 1. If “+” Then look for HIGH BP LOCATION with “-” sign.
 - 2. If found, Reference edge.



Reference edges Complete.

```
chr1 – only 1 DSB, both sides open:
(chr1, 179143829, '-') → TELOMERE
(chr1, 179143829, '+') → TELOMERE

chr21 – 3 DSBs sorted: 39,887,956 < 42,873,929 < 48,033,460

Consecutive pairs (rule: p_i '+' → p_{i+1} '-'):
(chr21, 39887956, '+') → (chr21, 42873929, '-') [segment between break 1 and break 2]
(chr21, 42873929, '-') → (chr21, 39887956, '+') [symmetric]

(chr21, 42873929, '+') → (chr21, 48033460, '-') [segment between break 2 and break 3]
(chr21, 48033460, '-') → (chr21, 42873929, '+') [symmetric]

Open ends (leftmost – and rightmost +):
(chr21, 39887956, '-') → TELOMERE [leftmost DSB, left side]
(chr21, 48033460, '+') → TELOMERE [rightmost DSB, right side]
```


#3 Cycle Traversal Using Compiled Data

Start Traversal

Sort the Rearrangement Edges in INCREASING ORDER.

(chr1, 179143829, '-'), (chr21, 39887956, '+'), (chr21, 42873929, '-'), (chr21, 48033460, '+')



Keep track of **VISITED NODES**. Traversal follows: **REAR** EDGE, **REF** EDGE, **REAR** EDGE, **REF** EDGE

↓
(chr1, 179143829, '-'), (chr21, 39887956, '+'), (chr21, 42873929, '-'), (chr21, 48033460, '+')

Traversal 1 — start: (chr1, 179143829, '-')

(chr1, 179143829, '-') –[REARR]→ (chr21, 48033460, '+') rearr_hops=1

(chr21, 48033460, '+') –[REF]→ **TELOMERE** → **OPEN PATH**

Visited: { (chr1, 179143829, '-'), (chr21, 48033460, '+') }

Start Traversal



~~(chr1, 179143829, '−')~~ , (chr21, 39887956, '+') , (chr21, 42873929, '−'), (chr21, 48033460, '+')

Traversal 2 — start: (chr21, 39887956, '+') ← not yet visited

(chr21, 39887956, '+') −[REARR]→ (chr21, 42873929, '−') rearr_hops=1

(chr21, 42873929, '−') −[REF]→ (chr21, 39887956, '+') ← start node, already visited

→ **CYCLE FOUND**, len(path)=2 → **C2**

Visited: { (chr1, 179143829, '−'), (chr21, 48033460, '+') , (chr21, 39887956, '+'), (chr21, 42873929, '−') }

Remaining nodes are skipped since all nodes are visited.



~~(chr1, 179143829, '−')~~ , ~~(chr21, 39887956, '+')~~, (chr21, 42873929, '−'), (chr21, 48033460, '+')

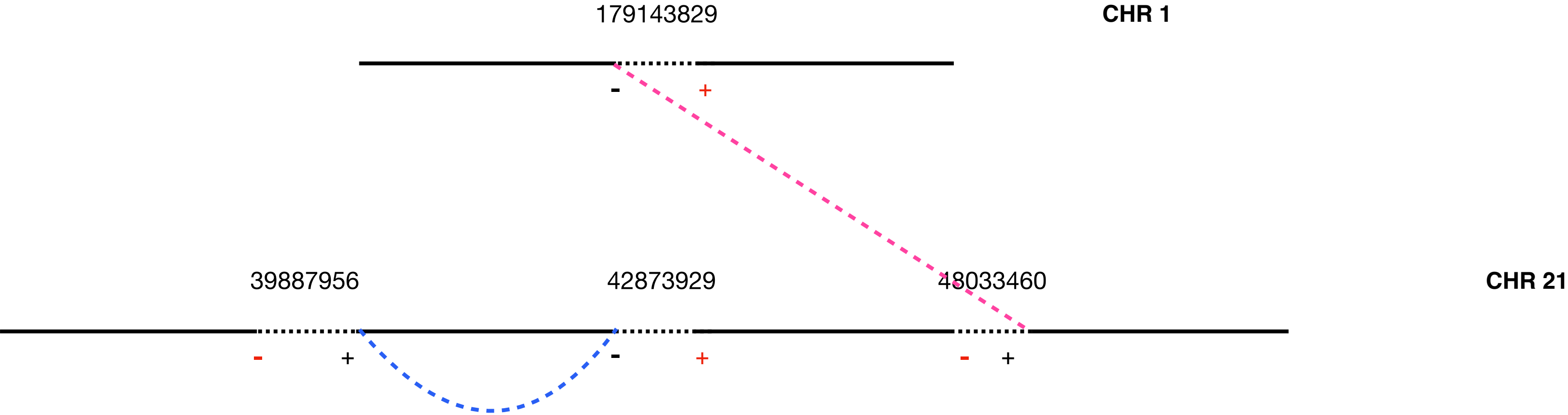


~~(chr1, 179143829, '−')~~ , ~~(chr21, 39887956, '+')~~ , ~~(chr21, 42873929, '−')~~ , (chr21, 48033460, '+')

~~(chr1, 179143829, '−')~~ , ~~(chr21, 39887956, '+')~~ , ~~(chr21, 42873929, '−')~~ , ~~(chr21, 48033460, '+')~~

Observed Traversal Complete.

Full ERG Chain (P08 - 501)



But wait ...

— TABLE C : MIXED matchings (3 / 3)
[hypothetical rejoins include both inter- and intra-chromosomal pairs]

Cycle structure	Count	%	
(none – open paths only)	2	66.7%	
C2	1	33.3%	← actual data
Total	3	100.0%	

The table shows C2, which means 2 DSBs... but we have 4 DSBs total (including 1 and 21) How?

That's because the observed path in the chain, does not include those other DSBs.

The C2 Chain, shown in the next slide, has involves only 2 DSBs. **Both** from **Chromosome 21**.

We also saw only 1 open path. Where is the second one coming from?

All possible combinations. The algorithm follows the traversal path once again but this time **without** the ACTUAL rearrangement edges. For each node, it maps the possible rearrangements and follows the traversal path (REAR , REF , REAR , REF,)

Possible Paths for ERG Chain

```
---
Matching 1 – actual data: {A↔D, B↔C}

A=(chr1, 179143829, '-') ↔ D=(chr21, 48033460, '+') [inter]
B=(chr21, 39887956, '+') ↔ C=(chr21, 42873929, '-') [intra chr21]
Traversal (same as actual):
A -[REARR]→ D -[REF]→ TELOMERE          → open path
B -[REARR]→ C -[REF]→ B                  → C2 ✓
Result: C2 ← actual data

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Matching 2: {A↔B, C↔D}

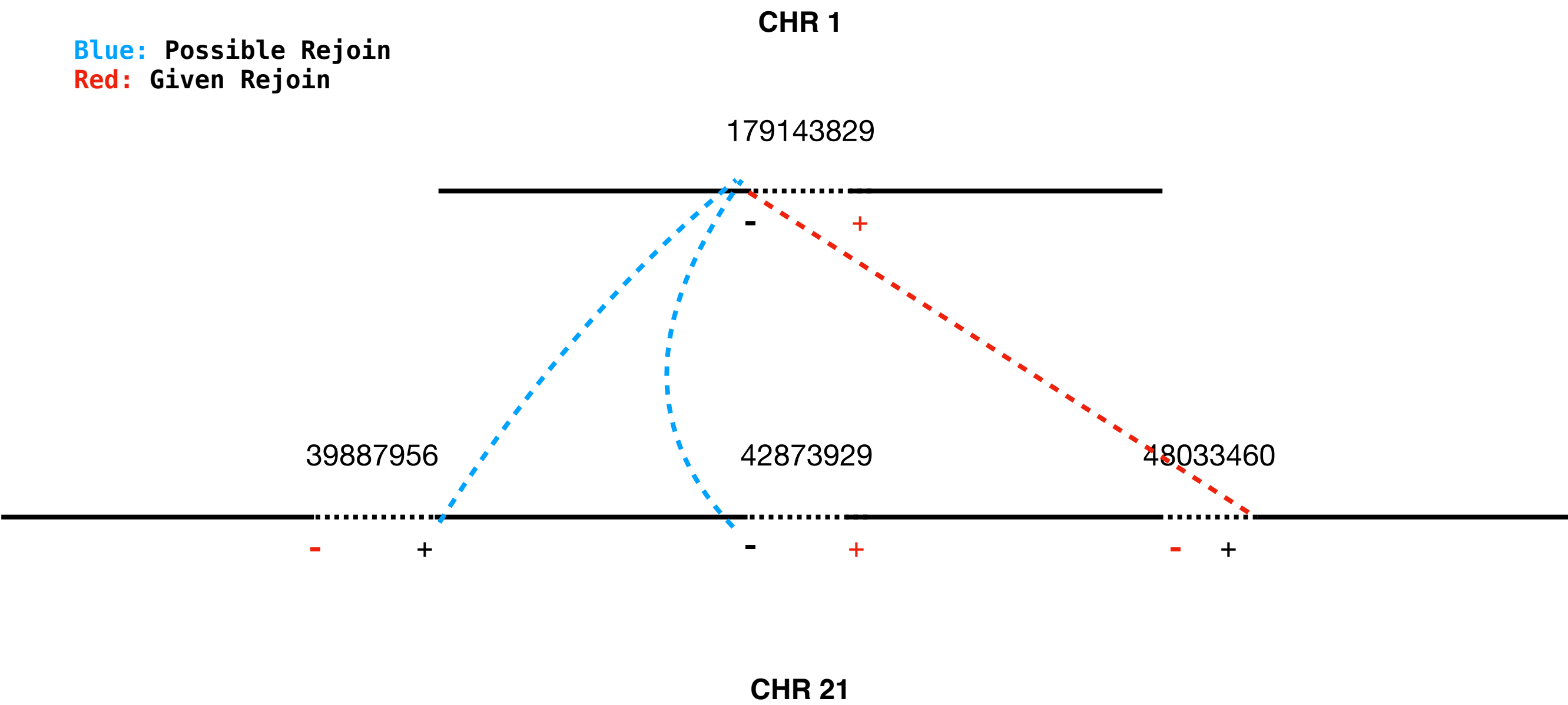
A=(chr1, 179143829, '-') ↔ B=(chr21, 39887956, '+') [inter]
C=(chr21, 42873929, '-') ↔ D=(chr21, 48033460, '+') [intra chr21]
Traversal starting at A:
A -[REARR]→ B=(chr21, 39887956, '+')      rearr_hops=1
B -[REF ]→ (chr21, 42873929, '-') = C
C -[REARR]→ D=(chr21, 48033460, '+')      rearr_hops=2
D -[REF ]→ TELOMERE                      → OPEN PATH
All 4 nodes visited in one open chain – no cycles.

Result: none (all open paths)

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Matching 3: {A↔C, B↔D}

A=(chr1, 179143829, '-') ↔ C=(chr21, 42873929, '-') [inter]
B=(chr21, 39887956, '+') ↔ D=(chr21, 48033460, '+') [intra chr21]
Traversal starting at A:
A -[REARR]→ C=(chr21, 42873929, '-')      rearr_hops=1
C -[REF ]→ (chr21, 39887956, '+') = B
B -[REARR]→ D=(chr21, 48033460, '+')      rearr_hops=2
D -[REF ]→ TELOMERE                      → OPEN PATH
Again one long open chain – no cycles.

Result: none (all open paths)
```

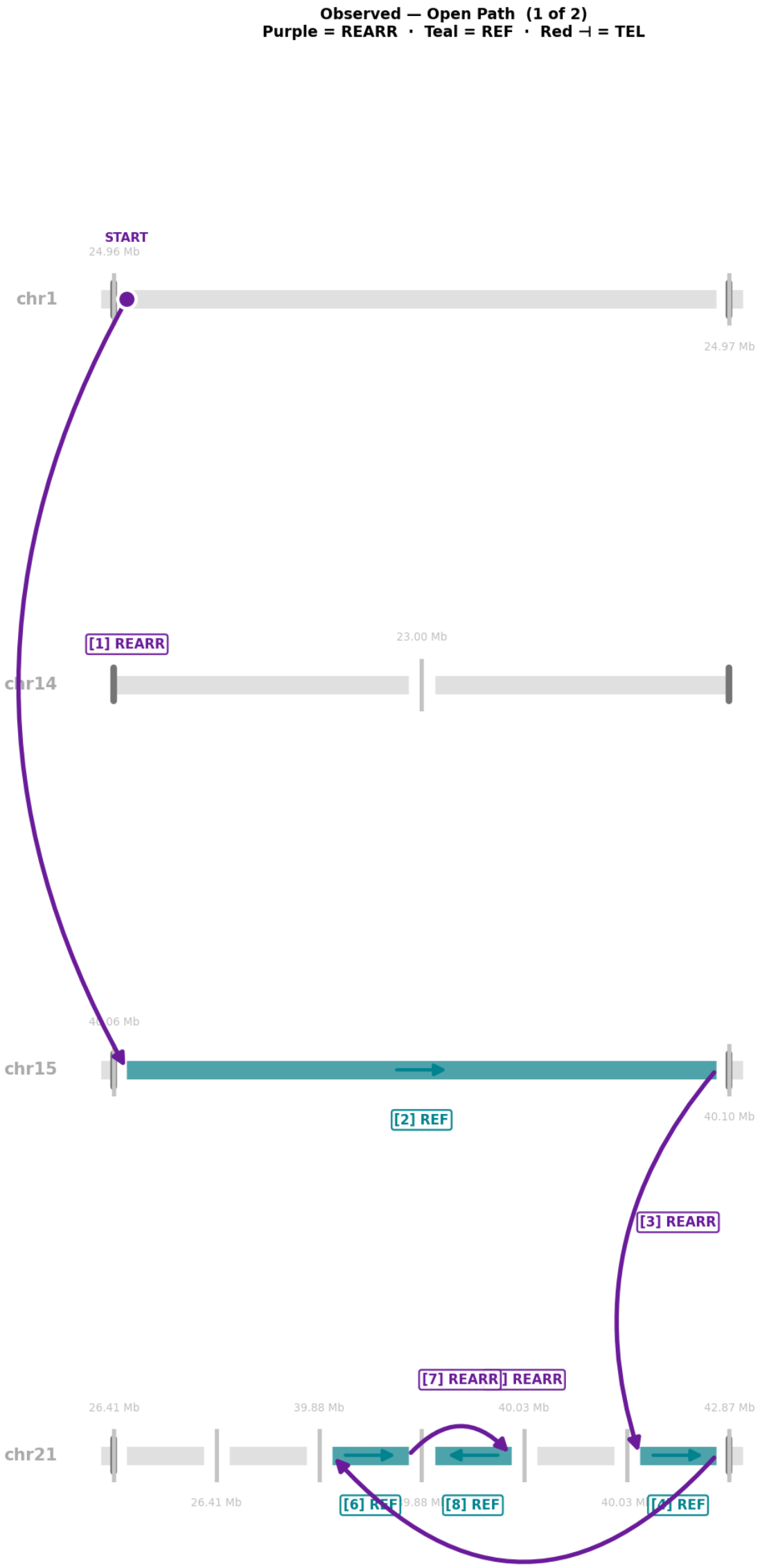
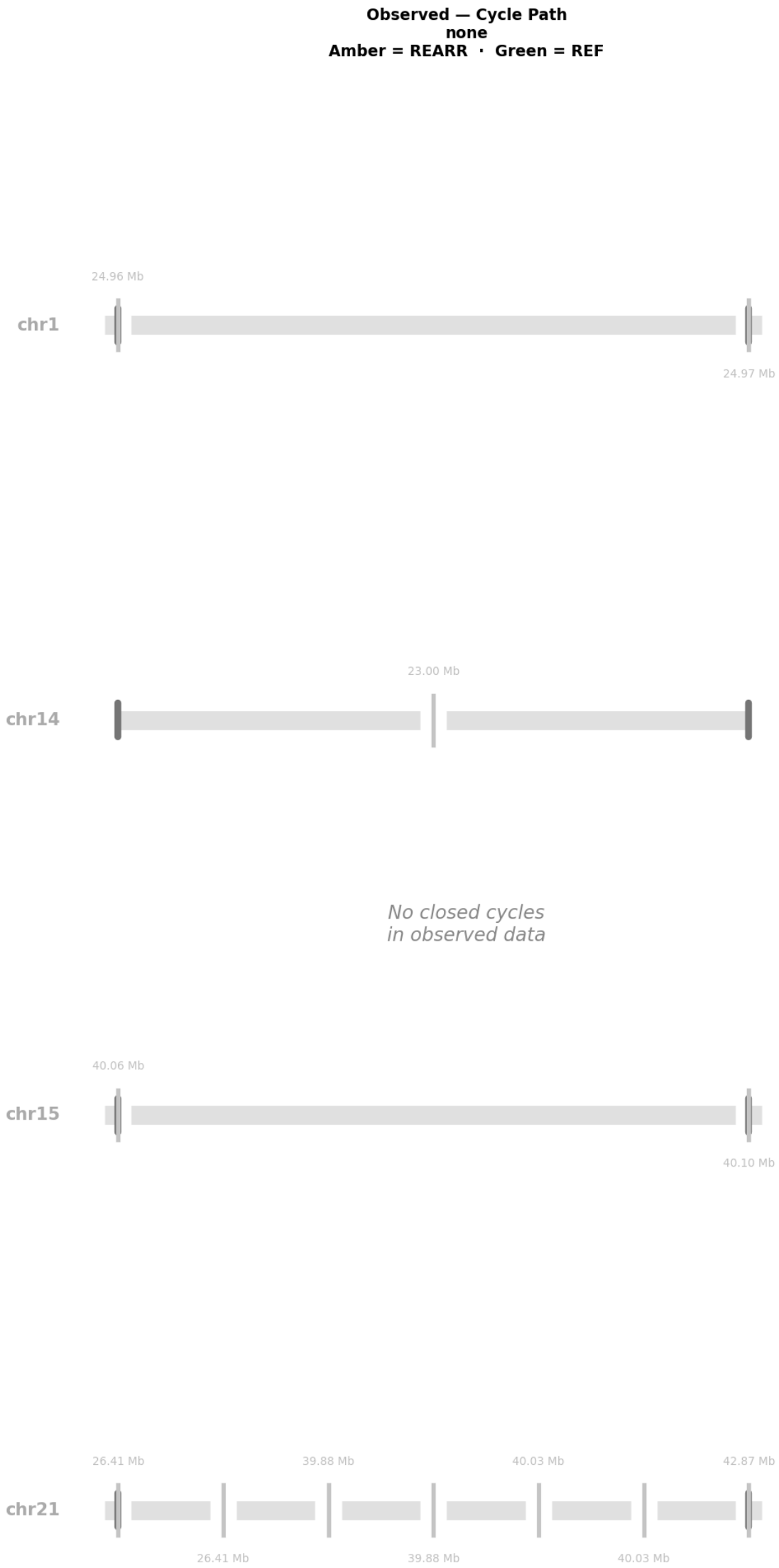
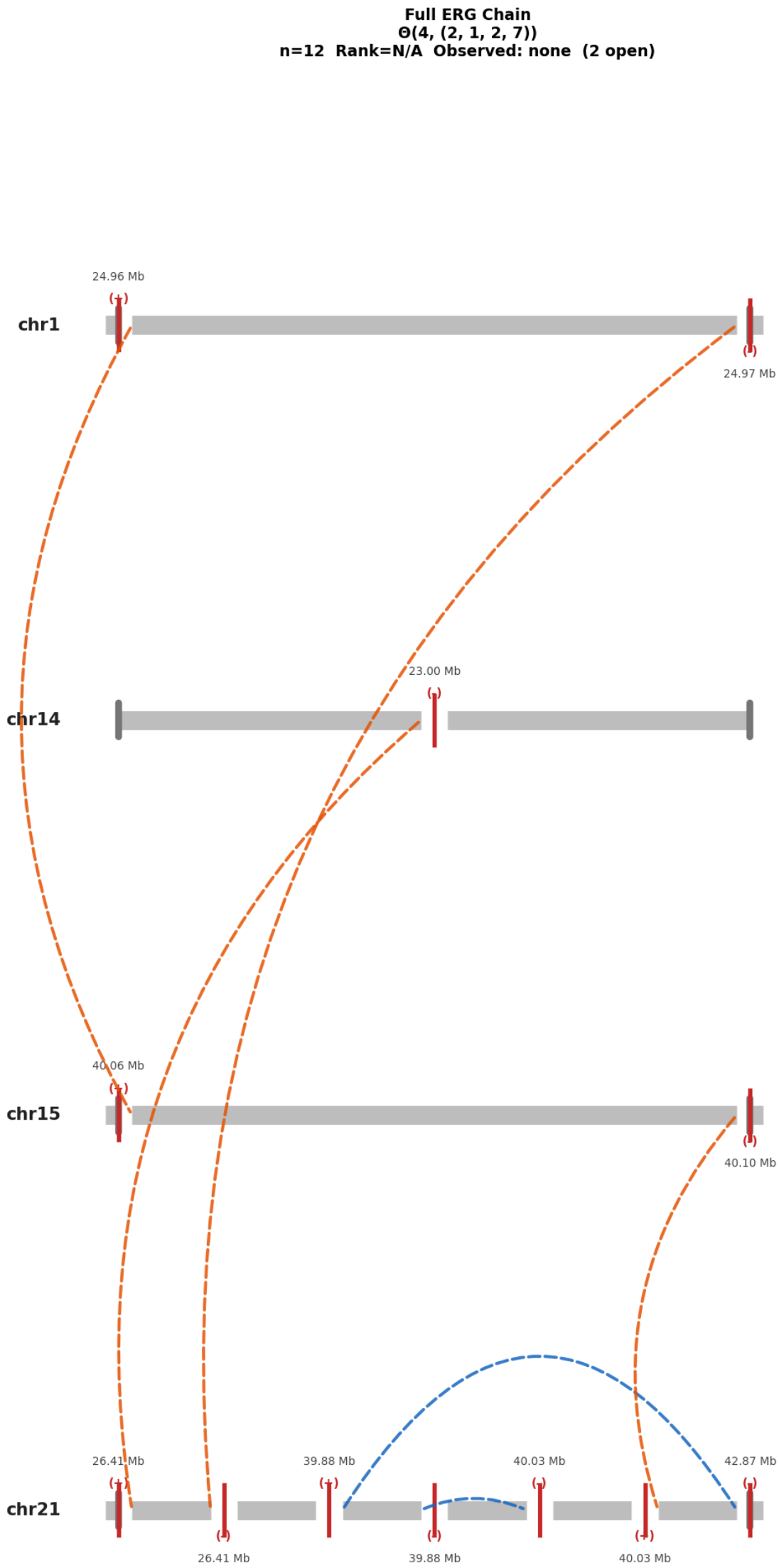


Notice we are only considering nodes (strands) that EXIST in the rows of the provided data.

The question this is answering is:

Given the same set of broken DNA ends, what if they rejoined differently? The "broken ends" are precisely the strand-specific endpoints that appear in the sequencing data — one per breakpoint per CSV row.

#4 Computing Cycle Diagram for Each Type of Cycle Possible for Each Patient



Intrachromosomal
Interchromosomal

REARR hop
REF segment
TELOMERE